NVPWRCONTROL / USER GUIDE

Control your GPU. Understand every step.

A practical English manual for NvpwrControl Stage11 Buildfix1 — first-time setup, every main tab, buttons, diagnostics, troubleshooting and recovery.

Audience: laptop users and volunteer testers with NVIDIA GeForce GPUs, especially RTX 50 Laptop devices.

Document basis: the actual WPF interface and button handlers in the Stage11 Buildfix1 full-source archive. Names match the English interface; experimental features may be disabled on individual drivers.

Build: Stage11 Integration / Buildfix1 (10 October 2026). This document describes available controls, not a certification that all operations work on every laptop.

> **READ THIS FIRST**
>
> NvpwrControl is experimental hardware-control software. Increasing power, current, voltage or clocks may cause crashes, overheating, instability or hardware damage. Use it on equipment you own or are authorized to test, and only within its verified capabilities.

YOUR SAFE FIRST SESSION

Detect hardware → save a baseline → validate one feature → make one small change → confirm live readback → restore if needed.

Start here — the 2-minute overview

NvpwrControl lets you inspect NVIDIA GPU power and clock capabilities, then **selectively** apply supported changes. It does not unlock every control automatically. A new GPU or driver may be read-only until its specific method has been verified.

The eight tabs

Power	TGP, persistent MAX/BASE, runtime PCF/F7, power readback and restore.
Tuning	Core, Memory, XBAR, supported clock/voltage offsets and extended-VRAM workflow.
Voltage	Separate Voltage Demand controls and NVVDD/MSVDD voltage offsets.
V/F Curve	Graphical core voltage/frequency point editing and a curve-profile lock.
Rails	NVVDD/MSVDD current-policy controls, persistent ceilings and recovery.
Telemetry ADVANCED TOOLS	Live sensor and policy values; read-only.
Compatibility ADVANCED TOOLS	Detected capabilities, validators, logs and exportable reports.
Settings	Language, colors, telemetry refresh interval.

Use **Advanced tools** in the top navigation bar to show **Telemetry** and **Compatibility**. The bottom **Details / Log** button opens the full status history in a separate window; the small status strip is intentionally compact.

Understand the four action types

READ / REFRESH	Inspects the GPU without an intentional overclock write.
VALIDATE	Tests whether a particular SET method is permitted. Some validators send a controlled unchanged-value or reversible hardware write.
RUNTIME APPLY	Changes the current driver session. It may be reset by driver reload/reboot or OEM policy.
PERSISTENT STAGE	Changes a tool-owned registry/VBIOS-shadow limit. Usually requires a reboot and fresh readback.

> RULE — A selectable slider is not proof that its requested value is effective. Always look for a successful operation result and the independent live value afterward.

1 Install and run on a new laptop

Requirements and preparation

- **Windows x64**, NVIDIA GPU and NVIDIA display driver. The project builds as a Windows WPF application targeting the .NET Framework 4.8 toolchain.
- For the **source archive**, use a Windows environment with the required .NET Framework compiler and WPF reference assemblies. The included **build.cmd** invokes **build.ps1**.
- The application asks for **Administrator** permission on launch because hardware/registry operations require elevation.
- Keep the laptop on reliable AC power, with working cooling and a stable stock configuration. Close other GPU tuners or undo their overrides before establishing a baseline.
- Make a backup of important data and keep a record of stock performance/temperature. Updating the NVIDIA driver can invalidate previously authorized methods.

If you received the source code

- **1.** Extract the **full** source ZIP to its own folder. Do not merge it with an older build.
- **2.** Run **build.cmd**; let every preflight and self-test finish. Never bypass an error or edit the approved SHA manifests to force a pass.
- **3.** After a successful build, locate **dist\NvpwrControl.exe**. The script also creates a runtime ZIP containing the EXE.
- **4.** Run the executable and approve Windows administrator elevation only if you trust the copy/source.
- **5.** If compilation fails, inspect **build-compiler.log**. The first CSxxxx error is usually the useful one.

If you received a compiled runtime ZIP

- **1.** Extract the runtime ZIP into a clean folder.
- **2.** Run **NvpwrControl.exe**; approve the UAC prompt if you trust the build.
- **3.** Wait for GPU detection and background deep compatibility to complete.
- **4.** Do **not** apply a saved overclock from another laptop. Start with read-only Compatibility and Power checks.

This release is distributed as experimental source/runtime content. Compatibility has not been proven on every RTX 5050/5060/5070/5070 Ti/5080/5090 Laptop, laptop OEM or NVIDIA driver. A detected profile is not a universal hardware certification.

First-run checklist

- **1.** Open **Power**. Check that the GPU identity, driver and VBIOS correspond to your own machine.
- **2.** Open **Advanced tools → Compatibility → Refresh**. Check that discovery has finished before testing a writer.
- **3.** Use **Export JSON** (read-only) and keep the original report outside the program folder.
- **4.** In **Power → Advanced / experimental power operations**, use **Capture baseline** before making persistent modifications, if available.
- **5.** Leave all offsets and current ceilings at stock until the relevant hardware validation has passed.

2 Power tab — what the controls mean

Think of NVIDIA power as layers: a persistent allowed **MAX**, a persistent **BASE**, a selected runtime request, the internal F7 source and the power the GPU actually draws. These numbers are not interchangeable.

Power sliders and modes

MAX TGP PERSISTENT	The requested **persistent upper limit** for graphics power. It is not an immediate power-consumption setting. Increasing it can require a registry/VBIOS-shadow stage and Windows reboot.
Base Limit PERSISTENT	A persistent board-power base value, separate from runtime TGP. Do not change it just to copy someone else's profile.
Manual BASE MODE	Use the selected Base Limit as an explicit desired value when the backend permits it.
Preserve OEM split MODE	Ask the backend to preserve the existing OEM base/boost relationship instead of manually forcing a base. A useful default on unfamiliar hardware.
Runtime TGP RUNTIME	Desired **current-session** wattage. Applying Runtime uses the PCF path plus native F7 where validated.

Everyday Power buttons

Apply coherent plan APPLY	Checks the selected MAX, BASE mode/base and Runtime values as one power plan. If persistent changes are necessary, the workflow may stage them and stop for a reboot instead of silently applying runtime values.
Apply Runtime (PCF + F7) RUNTIME WRITE	Applies the selected **Runtime TGP** through the existing validated PCF + E633/F7 route. It does **not** silently increase persistent MAX/BASE.
Review unified plan READ ONLY	Opens a **read-only preview** of selected targets and detected capability status. It does not issue a hardware SET.
Restore baseline RESTORE	Attempts to restore the saved **power** baseline and related owned power state. This is not the same as resetting every tuning, voltage or V/F offset.

Power readback — what to trust

Requested target	Your selected desired wattage, not necessarily what NVIDIA accepted.
PCF target	The upstream controller target.
Raw F7	The native NVIDIA power-source value read through the discovered contract.
Public CURRENT	What NVIDIA publicly reports as the current power limit.
GPU power draw	Actual observed consumption at that moment; can remain below a limit when the GPU is lightly loaded.

MAX ceiling	The present maximum recognized by NVIDIA; check after a persistent change/reboot.
Effective gap	Difference between requested and effective power; may indicate a clamp, not necessarily its cause.

3 Power — advanced actions and restart

The following controls are intentionally inside expandable sections. They are not routine first-day options.

Session clocks / persistent staging

Apply session: TGP + selected clocks MULTI-STEP WRITE	Applies runtime TGP and **checked Tuning rows** in a supervised session. It does not silently include persistent limits, Rails, Voltage Demand or V/F edits. Some mixed Memory configurations are not atomic and will be rejected.
Save MAX + BASE PERSISTENT	Stages owned persistent MAX and BASE changes, subject to semantic permissions and the selected BASE mode. Follow the reboot prompt when required.

Advanced / experimental power operations

Native F7 E633 probe EXPERIMENTAL WRITE	Uses the native F7 route without the normal PCF+F7 hold sequence. Intended for experienced testers only.
AD1/AD2 locked LOCKED	Research-only/native boost pathway. A “locked” label is intentional: arbitrary native boost SET is not established. Not a normal tuning option.
Native runtime A/B EXPERIMENTAL	Experimental Native Board Type0 route. Its SET contract can be rejected independently of PCF/F7; do not mistake this for a generic TGP failure.
Factory power policy RESTORE	Stages a return to the structurally resolved factory MAX/policy values. Persistent changes can require reboot.
Save BASE PERSISTENT	Stages a manual persistent BASE override when authorized.
Factory BASE RESTORE	Removes the tool-owned BASE override to return to the resolved factory base behavior.
Capture baseline SNAPSHOT	Captures the original power state and registry/ownership evidence for future recovery. Best used before your first persistent change.
Quarantine external overrides DESTRUCTIVE / REBOOT	Backs up and removes/replaces conflicting overrides created by **other** tools to reproduce the saved baseline. This can change external tool settings; use only if you understand the confirmation and backup.

Other Power options

Reassert effective CURRENT after under-load drift EXPERT	Optional rate-limited hold mechanism for PCF+E633 after drift. Not a guarantee against OEM/platform limits. Leave off during initial validation.
Experimental F7 load-transition probe EXPERT	Arms a read-only before/after comparison and at most one E633 reassert during a load transition. It is a diagnostic experiment, not an ordinary performance mode.
Restart later / Restart now REBOOT	Shown after a staged persistent operation. **Restart later** only dismisses the prompt; **Restart now** initiates Windows restart. Save your work first.

> IMPORTANT
>
> Raising MAX does not automatically raise CURRENT or actual GPU draw. A platform/OEM allocator may still limit GPU power or redistribute CPU/GPU budget. This build does not provide a verified universal OEM EC solution.

4 Tuning tab — clocks and memory

Each row has a **Use** checkbox, current/range information, a target control, units and a backend/status message. Only checked supported rows are included when you select **Apply selected**.

Tuning rows

Core clock offset RUNTIME	Adjusts the supported core-frequency offset in MHz. Increase gradually and test stability.
Memory clock offset RUNTIME / STAGED	Adjusts memory clock offset. The UI may show a project envelope up to **+6000 MHz**; that number is not permission to write above the live NVIDIA maximum and is not a stable recommendation.
XBAR clock offset RUNTIME	Adjusts a separate supported clock domain; availability depends on the NVIDIA driver/GPU contract.
NVVDD voltage offset RUNTIME	A voltage-related tuning row, not the same as NVVDD **current-policy** limits on Rails.
MSVDD voltage request RUNTIME	An independent request that may be unsupported or profile-gated.
GPC:XBAR ratio ×10000 EXPERT	Advanced ratio control expressed in raw UI units. Not a general-purpose beginner slider.

Tuning buttons

Refresh capabilities READ ONLY	Reads updated supported ranges and current tuning state.

Validate Full VF Memory VALIDATION	Checks the discovered full-VF memory binding/integrity contract. It is not a promise of +6000 MHz or a direct bypass of NVIDIA range limits.
Validate NVML Memory SET VALIDATION WRITE	Runs the dedicated NVML Memory SET validation, including its supported reversible checks. This may write and restore a small test offset.
Validate VRAM range mapping PERSISTENT / TWO REBOOTS	Stages a **small +100 MHz range-ceiling probe** (not a runtime overclock), then requires a reboot and independent NVML MAX verification. Completing proof requires restoring the probe and rebooting again.
Restore validation probe PERSISTENT RESTORE	Restores the tool-owned temporary VRAM-range validation probe. Follow the required reboot and verify stock public MAX afterward.
Restore VRAM range PERSISTENT RESTORE	Restores the owned persistent memory offset-range ceiling; it is not the same as resetting a runtime Memory offset.
Apply selected WRITE	Applies checked tuning rows. An extended memory target can instead trigger a protected persistent-range/reboot workflow if the current NVIDIA range is too small.
Reset tuning RESET	Resets supported tuning offsets using the existing tuning backend. Confirm current readback afterward.

5 Extended VRAM: understand the reboot workflow

Memory overclocking has **two distinct limits**: the memory offset NVIDIA currently accepts and the wider upper bound the program may be able to stage persistently after semantic VBIOS discovery. Selecting +4000, +5000 or +6000 is not the same as successfully applying it.

A typical supported-device workflow

- **1.** In **Compatibility** and **Tuning**, confirm that the GPU/driver/VBIOS identity is correct and that the Memory writer is supported.
- **2.** Run **Validate NVML Memory SET** when required. Stay within the current public NVIDIA range for initial functional checks.
- **3.** If extended-range discovery is **unvalidated** but eligible, run **Validate VRAM range mapping**. This is a special +100 MHz two-reboot causal test, not a +6000 MHz stress test.
- **4.** After the first reboot, open NvpwrControl and verify that the NVIDIA-reported maximum increased by exactly the expected amount. Use **Restore validation probe**, reboot again, and confirm exact return.
- **5.** Only if the resolver/validation authorizes it, select a moderate desired Memory target beyond the current public limit, check **Use**, and choose **Apply selected**. The program may stage the owned range ceiling and require a reboot.
- **6.** After the reboot, wait for fresh discovery. Verify the new live public Memory MAX. Then choose **Apply selected** again to apply the pending runtime Memory offset through the validated writer.
- **7.** Check actual memory frequency, stability and error-free behavior under a controlled load. If unstable, reset tuning and restore the owned VRAM range as required.

NOT AUTOMATIC	On startup the tool may restore a pending selection, but it does not automatically send an unapproved extended Memory SET. The follow-up Apply is initiated by the user.

Do not mix up these concepts

Memory clock offset	A session-level frequency adjustment.
Memory offset RANGE MAX	The allowed maximum offset reported by the NVIDIA driver; may require persistent staging.
Legacy Clock Programming MAX	A different absolute-frequency VBIOS record; not automatically the source of NVML offset-range authority.
Full-VF Memory	A private memory V/F configuration and integrity observation/validation route. It does not by itself prove effective overclock or stable +6000 operation.

6 Voltage tab — two different sets of controls

This tab contains independent categories. Do not confuse **Voltage Demand** for Core/XBAR/SYS/Video with NVVDD/MSVDD voltage offsets or the still-unproven Core Voltage Limits writer.

Voltage Demand: Core, XBAR, SYS, Video

When the driver exposes them, four rows appear with their current millivolt offsets, range, checkbox and target. The backend uses separate validation and recovery logic.

Validate Voltage Demand SET VALIDATION WRITE	Validates the separate signed NVIDIA Voltage Demand writer for the current GPU and driver. This may perform a reversible hardware probe.
Apply selected Demand WRITE	Applies only checked Voltage Demand rows (Core, XBAR, SYS or Video). Stage11 includes snapshot/readback and recovery handling for partial failures.
Reset Voltage Demand RESET	Requests reset of the four signed Voltage Demand offsets through their own backend.

NVVDD/MSVDD voltage-offset controls

Refresh voltage READ ONLY	Refreshes live voltage and tuning support information.
Apply selected voltage WRITE	Applies checked NVVDD/MSVDD voltage-offset rows through the tuning writer. These are independent from signed Voltage Demand.
Reset voltage offsets RESET	Resets eligible NVVDD/MSVDD voltage offsets to zero through the tuning writer.

Core Voltage Limits — MIN / MAX / OV

The “Core voltage limits” panel reports a detected read range, but a dedicated MIN/MAX/Overvoltage SET is **not** a confirmed universal writer in this build. If this section is locked or read-only, that is a capability limit, not a missing checkbox. Do not substitute another voltage slider to simulate it.

ONE CHANGE AT A TIME	Voltage changes can be riskier than simple clock offsets. On unfamiliar hardware, do not experiment with higher voltages/current ceilings until baseline, recovery and all relevant writer validations are in place.

7 V/F Curve tab — editing a curve

This page shows a graph of voltage/frequency points plus a table of points and offsets. The graph can be used for a visual profile adjustment; what the GPU actually accepts still depends on NVIDIA capability and readback.

How to edit

- **1.** Select a point on the graph.
- **2.** Drag the point vertically, or use the keyboard Up/Down arrows for the displayed adjustment step (15 MHz bins in this UI).
- **3.** Inspect the updated target in the graph/table.
- **4.** Use **Apply curve** when the backend is available, then **Refresh curve** and compare the returned data.
- **5.** Use **Restore curve** if you want to reapply the captured session baseline.

V/F Curve buttons

Refresh curve READ ONLY	Reads the current curve/point data again.
L — Lock selected EDITOR STATE	Toggles a **curve profile lock** around the selected point. The UI lock is **not** a proven low-level hardware V/F voltage lock.
Apply curve WRITE	Sends changed point offsets through the curve service when available.
Save profile SAVE FILE	Saves the curve points and visual lock metadata to a local text file; this does not make the profile universal for another GPU.
Restore curve RESTORE	Requests reapplication of the session baseline offsets captured by this UI. Confirm result and avoid assuming it resets unrelated controls.

Saved V/F profile location in this build: %LOCALAPPDATA%\NvpwrControl\state\vf-curve-profile.txt. The code stores point offsets and lock metadata, rather than an automatically portable multi-GPU profile.

8 Rails tab — NVVDD/MSVDD current policy

Rails controls NVIDIA **current-policy settings** for the NVVDD and MSVDD domains. It is not the Voltage tab, and changing a software current limit is not proof that the physical VRM overcurrent protection (OCP) threshold changed.

What you see

NVVDD / MSVDD Current Policy	Per-rail Use checkbox, currently applied current, default, current live maximum, target slider and backend/resolver status.
R616+ power-policy monitor	Read-only inventory of related NVIDIA policy records. Generic/sentinel records are not arbitrary controls.
Pending runtime replay	Indicates whether a persistent rail-ceiling change is awaiting a post-reboot runtime operation.

Rail buttons

Refresh rails READ ONLY	Updates rail discovery and compatibility information.
Validate Rail SET VALIDATION WRITE	Runs the dedicated current-policy writer validation, including an unchanged-value check and independent readback where supported.
Apply selected rail plan WRITE / POSSIBLE REBOOT	For checked rail rows, applies a protected plan. If target current exceeds the present live maximum, the system may need to stage a persistent ceiling first and require reboot.
Apply pending runtime RUNTIME WRITE	After a permitted ceiling stage and reboot, applies the stored runtime rail target when NVIDIA reports the new maximum and identity matches.
Reset runtime currents RESET	Returns eligible current-policy runtime settings toward the stored/default values through the rail backend.
Factory rail ceilings PERSISTENT RESTORE	Restores tool-owned persistent rail ceiling overrides to factory/baseline values, subject to reboot.
Export discovery JSON READ ONLY	Exports resolved capability/rail identity information to a JSON file. No hardware SET.

Why a rail control might be locked

The resolver must identify which policy record belongs to which rail; the same indices must **not** be assumed on a different GPU. A successful E61B request/readback does not automatically prove the bottleneck under load moved. If a writer is disabled, export the JSON and investigate instead of forcing the index.

RECOVERY	If a multi-rail operation is interrupted or reports partial failure, stop applying further limits and use the indicated recovery or read-only recovery verification. Do not delete the journal to clear an error.

9 Telemetry, Compatibility and logs

These tabs appear after clicking **Advanced tools**. Diagnostics are essential when adapting the program to another GPU, driver or laptop manufacturer.

Telemetry

Displays detected live values and units. It does not mean all desired limits have been applied. Missing or unsupported sensor values should be reported as unavailable, not guessed.

Compatibility — main buttons

Refresh READ ONLY	Runs a fresh capability check for the selected GPU and NVIDIA driver. A heavier deep discovery may finish in the background.
Capture diagnostic READ ONLY	Records a read-only NVIDIA power trace such as PCF, F7, public power and GPU draw. It does not change clocks or limits.
Export report EXPORT	Saves a human-readable diagnostic report; helpful for debugging.
Export JSON EXPORT	Saves a structured compatibility/discovery snapshot; the best starting point for reporting a new device.
Capture settings (read-only) READ ONLY	Expand to choose delay, trace duration and workload label. Optional raw PowerPolicy packets can be large and may contain vendor data. Optional CPU Package readings use an available sensor provider; no CPU power SET.
Quick 30s snapshot READ ONLY	Runs a convenient preset read-only capture with a 5-second start delay and 30-second trace.
Compare CPU/GPU logs (A/B CSV) READ ONLY	Compares two existing HWiNFO CSV captures offline. Select different baseline and test files.
Lenovo / NVPCF research snapshot READ ONLY	Lenovo-specific firmware/WMI diagnostic evidence collection; read-only and not required for universal NVIDIA tuning.
Export full debug JSON EXPORT	Detailed structured output for advanced support or developer analysis.

Shortcut: Ctrl+Shift+F9 saves a diagnostic trace while the diagnostics mode/hotkey is active, if another application has not already reserved that key combination.

Detailed status messages

Details / Log (bottom right): opens a separate scrollable diagnostic window with **Copy log** and **Close**. For a session history, expand **Power → Session operation log**. This replaces the older wall of text occupying the bottom of the main window.

10 Compatibility — advanced validation & recovery

Open **Advanced tools → Compatibility → Advanced diagnostics & hardware validation**. Unlike ordinary report export, the following validators can make controlled device writes. Only use the validator needed for your current task.

Validate PCF VALIDATION	Checks the runtime power-controller contract.
Validate Raw F7 SET VALIDATION WRITE	Validates the A632/E633/F7 SET path for the current GPU/driver/contract. It is independent from the Native Board Type0 experiment.
Validate Native Board SET EXPERIMENTAL	Research-only Type0 native board validation. A special control value of -1 and unsupported packets can be rejected; this must not be mistaken for a failure of F7.
Validate Voltage Demand SET VALIDATION WRITE	Hardware validation of the four signed demand-domain SET controls.
Validate Full VF Memory VALIDATION	Verifies the current Full-VF memory contract/identity and integrity expectations.
Validate NVML Memory SET VALIDATION WRITE	Per-device NVML memory offset SET/readback validation.
Validate Rail SET VALIDATION WRITE	Per-device NVIDIA Type19/E61B rail current-policy SET/readback validation.
Verify rail recovery (read-only) READ ONLY	Checks whether an existing rail recovery journal matches the original live state; never sends a hardware SET.
Verify Demand recovery (read-only) READ ONLY	Checks original Core/XBAR/SYS/Video demand state for the matching GPU/driver and clears only matching recovery evidence.
Auto-resolve VBIOS DISCOVERY	Attempts to obtain and identify the local VBIOS through the supported read-only tool/discovery route. May require administrator permissions.
Select ROM… DISCOVERY	Lets you select a VBIOS image manually for resolver analysis. Use only a ROM that belongs to the selected GPU.
Recover transaction RECOVERY	Attempts recovery of a previously interrupted power transaction using recorded trusted state. Follow any prompt; do not manually delete the journal.

Practical interpretation: “WRITE AUTHORIZED” means a particular driver method has passed its checks; it does **not** guarantee a larger TGP, a physically changed current limit or stable clocks. It is normal for some advanced methods to remain locked on a new driver.

11 Settings — make the interface comfortable

Language INTERFACE	Choose **English**, Russian or Ukrainian. This manual uses the English control labels.
Interface accent INTERFACE	Sets the active navigation/selection accent.
Background color INTERFACE	Changes the UI background color.
Button color INTERFACE	Changes button styling.
Slider color INTERFACE	Changes slider styling.
Telemetry interval (ms) INTERFACE	Controls the ordinary sensor refresh interval. Very aggressive intervals may add diagnostic overhead.
Apply colors INTERFACE	Applies the selected palette to the running UI.
Save settings SAVE	Saves language, appearance and telemetry preferences. This is **not** an overclock profile-save button.

The WPF shell is designed for scaled Windows displays, including the documented 100–200% scaling use case. Adjust window size or Windows display scaling if controls feel crowded.

Your normal everyday workflow

- **1.** Open NvpwrControl and let compatibility detection complete.
- **2.** Confirm **Power** live values and baseline. Do not carry power targets over from a different device.
- **3.** For runtime power only, select a supported target within the live MAX and click **Apply Runtime (PCF + F7)**.
- **4.** For clocks only, go to **Tuning**, check the row(s) in **Use**, set supported targets and click **Apply selected**.
- **5.** For a combination of runtime TGP and supported clocks, use **Apply session: TGP + selected clocks**, respecting its narrower atomicity rules.
- **6.** If you need higher persistent MAX/BASE, use **Review unified plan** first, then **Apply coherent plan** and complete any requested reboot.
- **7.** After each change, read the resulting telemetry and test stability in a controlled workload. Do not treat an accepted request as confirmed performance.
- **8.** At the first sign of instability or unexplained limits, stop; use the corresponding Reset/Restore and recovery process.

12 Troubleshooting, restoration and support

Common messages

VALIDATION_REQUIRED / READ_ONLY	The current device/driver method has not been validated for writing. Run only the appropriate validator; if it remains locked, export JSON.
F7_VALIDATE_SEMANTIC_CONTRACT	The proof of the A632/E633 method shape/flow is incomplete or stale. Refresh full discovery and revalidate; do not bypass the guard.
NATIVE_BOARD_NOOP_REJECTED / rc=-1	Experimental Native Board Type0 unchanged-value request was rejected. This is separate from PCF+F7 and does not mean all TGP control is broken.
BOARD_CONTROL_SENTINEL	A read value such as -1 is not a normal watt setting. The validator should avoid replaying it as a real power target.
REBOOT_REQUIRED	A persistent limit was staged but is not confirmed live. Save work, reboot, run discovery and check readback before continuing.
PARTIAL_EFFECTIVE / CLAMPED	One or more requests did not become fully effective. Compare Requested, PCF, F7, public CURRENT, GPU draw and temperature; the reason may remain unknown.
RECOVERY_PENDING	A previous transaction needs verification/recovery. Do not start unrelated writes or remove state files.
PRE-FLIGHT FAILED / csc failed	Build/package error, not a GPU error. Inspect the earliest preflight failure or build-compiler.log CSxxxx line. Never disable checks or alter SHA files merely to compile.

Restore safely — choose the right control

- **Runtime TGP:** use the validated power Restore baseline operation if you captured one.
- **Clock offsets:** use Tuning → Reset tuning; check all affected domains afterward.
- **Voltage Demand:** use Voltage → Reset Voltage Demand and recovery verification if needed.
- **NVVDD/MSVDD voltage offsets:** use Voltage → Reset voltage offsets.
- **Current policy:** use Rails → Reset runtime currents; for tool-owned ceilings, use Factory rail ceilings and reboot if required.
- **Extended VRAM:** use Restore validation probe/Restore VRAM range as appropriate and perform the required reboot/readback.
- **V/F Curve:** use Restore curve for the session curve baseline, then check current values.

Do not assume **Restore baseline** in Power is an atomic reset of every power, rail, voltage, memory and V/F state. Recovery is subsystem-specific.

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13 First-time tester checklist for a new GPU

If you are distributing this application to testers, ask them to follow this sequence before reporting a feature as supported. It prevents a known-good Lenovo value or index from being mistaken for a universal one.

- **1.** Run the compiled x64 executable on the tester's own GPU. Record the exact GPU, VBIOS, NVIDIA driver and laptop OEM.
- **2.** Open **Advanced tools → Compatibility** and wait for both fast and deep discovery. Do not run an overclock while it is still resolving.
- **3.** Click **Export JSON** and keep a copy. Check the capability matrix: writable, read-only, unsupported or validation required.
- **4.** On stock settings, use **Capture baseline** and optionally **Export report**. Keep those originals for recovery.
- **5.** Run only the specific validation required for one intended feature; observe the outcome and any exact readback. Avoid changing multiple independent domains during initial testing.
- **6.** If a validator fails, stop that feature and export **full debug JSON**. Do not borrow a rail ordinal, registry address, RVA or power value from another laptop.
- **7.** After a permitted, small runtime change, capture **Quick 30s snapshot** or a longer diagnostic before/under load. Compare requested vs effective values.
- **8.** After a staged persistent change, reboot and re-run discovery. Verify the new allowed range; separately apply the intended runtime request.
- **9.** Before changing the NVIDIA driver or switching GPU, restore tool-owned persistent changes and save your logs. Revalidate on the new identity.

What to include in a useful issue report

- App archive/build name and the exact action/button used.
- Exact GPU model, driver version, VBIOS version and laptop model.
- Compatibility JSON, plus readable report or relevant log excerpt.
- The first error code/message, before later follow-on errors.
- What was **Requested**, what NVIDIA returned on readback, and the current MAX/temperatures.
- Whether the system had to reboot, and whether another tuning tool had persistent overrides.

PRIVACY	Diagnostic exports can contain driver paths, device identifiers, firmware details or vendor-specific raw buffers. Review them before publishing publicly. Share raw private packet captures only with people you trust.

Current scope and what is still experimental

Stage11 Buildfix1 includes working branches inherited from prior tests, but **universal A619 runtime STATUS transport, independent Core Voltage Limits MIN/MAX/OV SET, genuine hardware V/F lock, and fully verified all-device atomic MaximumBatch** are not completed universal features. OEM EC / platform-entitlement writing and CPU power-limit writes are outside this release.

14 Quick glossary

BASE	Persistent board-power base policy level.
MAX	Persistent maximum allowed policy setting; not current GPU draw.

Runtime TGP / CURRENT	Current-session power request or effective publicly reported cap, depending on the telemetry field.
PCF	NVIDIA-related runtime power-controller path used by NvpwrControl.
F7 / A632 / E633	Native source-readback and SET method family for GPU power.
NVML	NVIDIA Management Library; includes public reported power and memory-offset controls.
VBIOS / ROM	The GPU firmware image and its policy/clock tables.
romOverride	Tool-owned persistent registry shadow/override mechanism used for particular limits.
Rail / Type19 / E61B	NVIDIA current-policy records and writer route for identified NVVDD/MSVDD rails.
Voltage Demand	Separate signed per-domain voltage requests for Core/XBAR/SYS/Video.
V/F Curve	Voltage/frequency point data and offsets. Curve-profile lock is not verified hardware voltage locking.
Resolver	Dynamic discovery of a driver/VBIOS-specific function, structure or record; discovery alone does not authorize a write.
Readback	A fresh query showing what the driver reports after an attempted change.
Recovery journal	Stored metadata used to handle an interrupted/partial transaction, kept outside the EXE in local application data.
Clamp	A limit or controller that reduces the effective result. The source must be supported by evidence, not inferred from one reading.

Final reminder: A safe first-time user should spend more time reading capability statuses and verifying a small change than trying the highest available slider position. An unsupported writer staying locked is the correct behavior on an unknown driver or GPU.

Documentation provenance: verified against src/WpfMainWindow.cs, src/Program.cs, build.ps1, src/StatePaths.cs, README_EN.md and Stage11 Buildfix1 release notes from the supplied full-source distribution. This document does not certify hardware outcomes that were not independently tested.